Find the maximum n for which there exists a graph of metric dimension k that contains a subgraph isomorphic to the n-dimensional hypercube Q_n.

START

Post 2 · topic 2205179

A hint for getting started (a similar idea might help for some of the related problems): By thm 2.2 in my paper (resource 4), we have: $2^n \leq (n+1)^k$.

PROGRESS

Post 3 · topic 2205179

Is $n = \Theta(k \log k)$? Anyone see how to get the upper/lower bounds?

PROOF

RESULT

PUBLISHED

Post 4 · topic 2205179 arXiv 2008.13302, Thm 3.2

Upper bound: $n \leq k \log_2(n+1)$ implies $n \leq 3k \log_2(k)$.
Lower bound: $Q_{\frac{1}{2} k \log_2(k)}$.
Anyone see what should be the coefficient of the $k \log_2(k)$?

QUESTION

Post 6 · topic 2205179

By the way, what is the construction for the lower bound?

ANSWER

Post 7 · topic 2205179 · answers Post 6

Good question, the construction for the lower bound for this problem is the landmark placement for $Q_{\frac{1}{2} k \log_2(k)}$ that gives the upper bound for $\dim(Q_n) = 2\frac{n}{\log_2(n)}(1 \pm o(1))$. This paper gives a landmark placement (the set of landmarks for this problem is the set of subsets that they construct in the paper): Cantor & Mills — *Determination of a subset from certain combinatorial properties* (1966).